COA_Unit-1: Syllabus

Basic Structure of Computers

- Basic organization of computers
- Historical perspective
- Bus structures

Data Representation

- Data types – Number Systems, ASCII Codes
- Complements
- Fixed point representation
- Floating point representation
- other binary codes
- error detection codes

Computer Arithmetic

- Addition and subtraction
- Multiplication algorithms
- Division algorithms

COA_UNIT - III

Central Processing Unit:

- General Register Organization
- Stack Organization
- Instruction Formats
- Addressing Modes
- Data Transfer and Manipulation
- Program Control
- Reduced Instruction Set Computer

Microprogrammed Control:

- Control Memory
- Address Sequencing
- Design of Control Unit
- Microprogram Example

UNIT-4

PART-A: Memory Organization

- Memory Hierarchy
- Main memory
- Auxiliary memory
- Associate Memory
- Cache Memory
- Virtual memory

UNIT-4

PART-B :I/O Organization

- Peripheral Devices
- I/O interface
- Asynchronous data transfer
- Modes of transfer
  - Programmed I/O
  - Priority Interrupt
  - Direct memory access

UNIT-V: Multiprocessors & Pipelining

PART-A: Multi-Processors

- Characteristics of Multiprocessors
- Interconnection Structures
- Inter Processor Arbitration

PART-b: Pipelining

- Parallel Processing
- Pipelining
- Instruction Pipeline
- RISC Pipeline
- Array Processor

2. Register Transfel Language & Micro Operation

- Register transfer language (RTL)
- Register transfer
- Bus transfer
- Memory transfer.
- Arithmatic micro-operations.
- Logical micro-operations.
- Shift micro-operations
- Arithmetic-logic-shift unit.